

Healthcare Operations Analytics Report

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Executive Summary

This project analyzes over 9,000 patient encounters from April 2023 to October 2024 to understand how patients move through the hospital system and where operational friction slows them down. With my background in respiratory care and healthcare analytics, I approached this dataset with both clinical intuition and analytical structure. The goal was simple: identify the operational patterns that shape patient experience and highlight opportunities to improve flow, reduce delays, and support care teams.

Dataset Overview

-  Demographics — Age, gender, race
-  Visit Details — Arrival time, visit type
-  Departments — Referral location
-  Wait Times — Minutes from arrival to service
-  Satisfaction — Patient-reported experience
-  Providers — Doctor ID linkage

The dataset covered a full 15-month period, allowing for trend analysis and operational pattern recognition.

Data Preparation

To ensure accuracy and consistency, I performed several key steps:

- Standardized date/time formats
- Cleaned inconsistent gender entries

- Replaced missing department values with a neutral placeholder
- Preserved null satisfaction scores to avoid bias
- Validated encounter ranges and structural integrity

These steps created a reliable foundation for deeper operational analysis.

Key Findings

Patient Volume

Steady month-to-month volume with clear year-over-year growth — a sign of predictable demand rather than sudden surges.

Visit Types

Emergency visits were slightly more common, reinforcing the need for flexible staffing and rapid-response workflows.

Wait Times

Wait times were consistently elevated across departments, indicating a system-wide issue rather than isolated inefficiencies.

Bottlenecks

Certain department-hour combinations (e.g., evenings in specific specialties) showed disproportionately high delays — the kind of bottlenecks that often go unnoticed without data.

Patient Satisfaction

Scores were generally low and consistent across departments, suggesting that operational delays — not clinical care — are driving dissatisfaction.

Shift Load

Night shift physicians handled more patients than other shifts, revealing a misalignment between staffing and actual demand.

💡 Insights That Matter

- Delays are systemic, not department-specific
 - Bottlenecks cluster at predictable department-hour intersections
 - Staffing patterns do not fully reflect patient arrival patterns
 - Satisfaction is strongly influenced by operational flow
 - Emergency visits remain the most unpredictable segment
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🚀 Recommendations

1. Streamline Intake & Triage

Reducing friction at the front end can significantly lower wait times without adding staff.

2. Target High-Impact Bottlenecks

Focus on department-hour intersections where delays are most severe — small changes here create outsized impact.

3. Align Staffing With Demand

Shift-level adjustments can balance workloads and reduce strain on night shift clinicians.

4. Improve Efficiency Before Expanding Capacity

The data suggests inefficiencies — not demand spikes — are driving delays. Process improvements should come first.

🧠 Personal Reflection

This project reminded me of why I moved from clinical care into analytics. Data gives us the ability to see patterns that aren't visible on the hospital floor — patterns that directly affect patients and the clinicians caring for them. Behind every wait time is a person who is anxious or uncomfortable. Behind every staffing imbalance is a clinician doing their best under pressure. My goal is to continue bridging clinical insight with data-driven decision-making to create better experiences for both.